SNIGDHASHREE MALLICK

International Institute of Information Technology, Bangalore

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Education

International Institute of Information Technology, Bangalore — **Sep. 2021 – Present**
PhD in Computational Fluid Dynamics — *3.53/4*

- Supervisor : Prof. Shiva Kumar Malapaka
- Co-supervisor : Prof. Amit Chattopadhyay

Fakir Mohan University — **Aug. 2015 – Jul. 2017**
M.Sc. in Physics (Specialization in Nuclear and Particle Physics) — *8.5/10*

Bhadrak Autonomous College (Fakir Mohan University) — **Aug. 2012 – Jul. 2015**
B.Sc. in Physics (Hons.) — *74.8%*

Kalinga Bharati Residential College,Cuttack — **Aug. 2010 – Jun. 2012**
Intermediate (Physics, Chemistry, Math, IT) — *75.6%*

Chandbali Girls' High School, Chandbali — **Jun. 2007 – Jun. 2010**
High School — *90.83%*

Experience

Guest Faculty — **Sep. 2018 – Jul. 2019**
Chandbali College — *Chandbali, Odisha*

- **Delivered** lectures for the subject *Digital Electronics* and *Heat and Thermodynamics*
- Graded around 70 assignments per grading period

Project

Saha Institute Of Nuclear Physics — **May 2017 – Jun. 2017**
Summer intern (PI:Prof. Anjali Mukherjee) — *Bidhannagar, Kolkata*

- **Explored** the significance of $^{12}\mathrm{C} + ^{13}\mathrm{C}$ and $^{13}\mathrm{C} + ^{13}\mathrm{C}$ fusion reactions in astrophysical scenario.
- **Measured** the $^{12}\mathrm{C} + ^{12}\mathrm{C}$ fusion cross sections at energies reaching the Gamow peak region.

Mentorship

Teaching Assistant | *Introduction to Astronomy and Astrophysics at IIIT Bangalore*

- Served as a **Teaching Assistant** for the course in **two semesters** (Aug–Dec 2022 and Aug–Dec 2023).
- Graded 100+ assignments per grading period to aid the teacher.

Technical Skills

- **Programming Languages:** C, Python, Fortran
- **Scientific Data Analysis Tools:** Topology Toolkit (TTK), Gudhi, Ripser, VisIt, ParaView
- **Relevant Courses** : High Performance Computing, Computational Topology, Advanced Mathematics

Manuscript(s) Under Preparation

- Mallick S., Ramamurthi Y., Malapaka, S.K., and Chattopadhyay, A., *Wasserstein Distance and the Spatio-Temporal Localization of strong fluctuations in 3D Helically Rotating Hydrodynamic Turbulent Flows* (final title may vary)

Achievements

- Awarded the **Prateeksha Scholarship** for a five-year duration commencing in September 2021 to pursue PhD at IIIT,Bangalore.
- Successfully **completed** the State of the Art Seminar for PhD in November 2024.
- Successfully **completed** the Comprehensive Exam for PhD in September 2023.
- **Granted** national-level scholarships: National Rural Talent Search (NRTS) in 2008 and NMMS (National Merit-cum-Means Scholarship) in 2009.

Conferences/Workshops

- **Attended** OneMath World School: Topological Data Analysis and Applications at Chennai Mathematical Institute (CMI) (Feb 20-28, 2026).
- **Attended and presented** a talk titled *Topological Data Analysis of Low-Resolution 3D Helically Rotating Hydrodynamic Turbulent Flows* at the Nordita Winter School on *Cosmological Magnetic Fields: Generation, Observation, and Modeling* (Jan 11–23, 2026), Nordita, Stockholm.
- **Attended** Bangalore Vis Workshop (Nov 22, 2025) at IIIT Bangalore.
- **Presented** the poster *Can we localize Intermittency in both space and time?* at the RISE Open House, IIIT Bangalore (Feb 22, 2025).
- **Attended** Bangalore Vis Workshop (Sep 29, 2023, online).
- **Moderated** the workshop on *Various Aspects of Quantum Computing* organized by the IIITB COMET Foundation (May 19, 2023).

Miscellaneous

- **Attended** Inter-University Centre for Astronomy and Astrophysics-National Centre for Radio Astrophysics (IUCAA-NCRA) Grad school (Oct 2021-Feb 2022).